

Differential Equations

Separable Differential Equations

- We examine a subclass of linear and nonlinear first order equations. Consider the first order equation

$$\frac{dy}{dx} = f(x, y)$$

- We can rewrite this in the form

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0$$

- In differential form,

$$M(x, y)dx + N(x, y)dy = 0$$

- If M is a function of x only and N is a function of y only, then

$$M(x)dx + N(y)dy = 0$$

- In this case, the equation is called **separable**.

Example 1:

- Solve the following first order nonlinear equation:

$$\frac{dy}{dx} = \frac{x^2 + 1}{y^2 - 1}$$

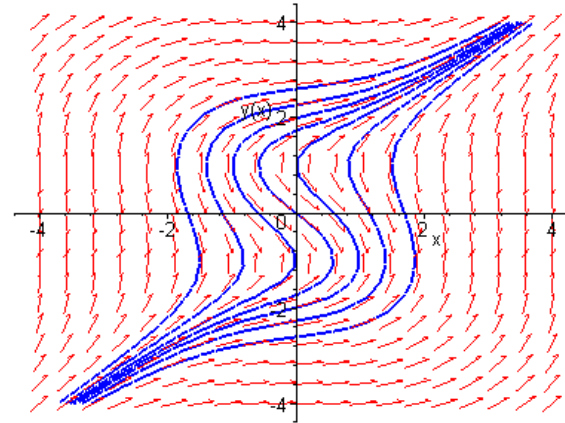
- Separating variables, and using calculus, we obtain

$$(y^2 - 1)dy = (x^2 + 1)dx$$

$$\int (y^2 - 1)dy = \int (x^2 + 1)dx$$

$$\frac{1}{3}y^3 - y = \frac{1}{3}x^3 + x + C$$

$$y^3 - 3y = x^3 + 3x + C$$



- The equation above defines the solution y implicitly. A graph showing the direction field and implicit plots of several **integral curves** for the differential equation is given above.

Example 2:

- Solve the following first order nonlinear equation:

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}$$

- Separating variables and using calculus, we obtain

$$2(y-1)dy = (3x^2 + 4x + 2)dx$$

$$2 \int (y-1)dy = \int (3x^2 + 4x + 2)dx$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + C$$

Example 3:

- Consider the following initial value problem:

$$y' = \frac{y \cos x}{1 + 3y^3}, \quad y(0) = 1$$

- Separating variables and using calculus, we obtain

- Using the initial condition, it follows that

A **separable** differential equation is one that can be written so that the independent variable terms (along with its differential) are collected to one side of the equal sign, and the dependent variable terms (and its differential) to the other.

Example: $y' = xy^2$ is separable. It is first written as $\frac{dy}{dx} = xy^2$, then “separated”:

$$\frac{dy}{y^2} = x \, dx.$$

To solve a separable differential equation, integrate both sides and solve for y (if possible). For example, we had $y' = xy^2$. This is separated as $\frac{dy}{y^2} = x dx$.

$$\int \frac{dy}{y^2} = \int x dx \text{Integrate both sides.}$$

$$-\frac{1}{y} = \frac{1}{2}x^2 + C \text{Don't forget the constant of integration.}$$

$$\frac{1}{y} = C - \frac{1}{2}x^2 \text{Negate. The } C \text{ "absorbs" the negative.}$$

$$y = \frac{1}{C - \frac{1}{2}x^2} = \frac{2}{C - x^2} \text{Solve for } y. \text{ Note that } 2C \text{ is written as } C.$$

- The constant of integration C is just a generic constant at this point. It absorbs all constants that come near it, so to speak. For example, $e^C = C$, $-C = C$, $2C = C$, $\frac{1}{C} = C$, and so on.
- The C can be determined with an initial condition. For example, suppose we have $y' = x + xy$ with $y(0) = 3$. The general solution is $y = C^{0.5x^2} - 1$. To find C , let $x = 0$ and $y = 3$:

$$3 = Ce^{0.5(0)^2} - 1 \quad \rightarrow \quad 3 = C - 1 \quad \rightarrow \quad C = 4.$$

Thus, the particular solution is $y = 4e^{0.5x^2} - 1$.

Example: the rate of change in the value of a stock is inversely-proportional to the square of the value of that stock. If the stock's value was \$20 at noon, and was \$23 at 3 p.m., what is the stock's value at 5 p.m.?

$$V'(t) = \frac{k}{[V(t)]^2} \quad \rightarrow \quad \frac{dV}{dt} = \frac{k}{V^2}$$

$$V^2 dV = k dt \quad \rightarrow \quad \int V^2 dV = \int k dt$$

$$\frac{1}{3} V^3 = kt + C$$

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$$\begin{aligned}\frac{1}{3}V^3 &= kt + C \\ V^3 &= 3kt + C \\ V(t) &= \sqrt[3]{3kt + C}.\end{aligned}$$

To find C , use $(0,20)$, where $t = 0$ is noon: $20 = \sqrt[3]{3k(0) + C} \rightarrow 20^3 = C$

Thus, we have $V(t) = \sqrt[3]{3kt + 8000}$.

To find k , use $(3,23)$: $23 = \sqrt[3]{3k(3) + 8000} \rightarrow 23^3 = 9k + 8000$

$$23^3 - 8000 = 9k \rightarrow 4167 = 9k \rightarrow k = \frac{4167}{9} = 463.$$

The particular solution is $V(t) = \sqrt[3]{3(463)t + 8000}$ or $V(t) = \sqrt[3]{1389t + 8000}$.

At 5 p.m., the stock's value is $V(5) = \sqrt[3]{1389(5) + 8000} \approx \24.63 .

Linear Differential Equations

There is a process by which most first-order linear differential equations can be solved. This uses an **integration factor**, denoted $\mu(x)$ (Greek letter “mu”).

The differential equation must be in the form

$$y' + f(x)y = g(x).$$

To find $\mu(x)$, we perform the following process (next slide)

Starting with $y' + f(x)y = g(x)$, multiply both sides by $\mu(x)$:

$$\mu(x)y' + \mu(x)f(x)y = \mu(x)g(x)$$

The left side is a product-rule derivative of $(\mu(x)y)$:

$$(\mu(x)y)' = \mu(x)y' + \mu'(x)y.$$

Thus, we have $\mu(x)y' + \mu'(x)y = \mu(x)y' + \mu(x)f(x)y$.

This forces $\mu'(x) = \mu(x)f(x)$. *(next slide)*

Now we find $\mu(x)$. From the last slide, we had $\mu'(x) = \mu(x)f(x)$.

This is a separable differential equation... so separate:

$$\frac{d\mu}{\mu(x)} = f(x)dx.$$

Integrating both sides, we have

$$\int \frac{d\mu}{\mu(x)} = \int f(x)dx .$$

(next slide)

After integration, we have

$$\ln \mu(x) = \int f(x) dx + C.$$

Here, we only need one form of the antiderivative, so we let $C = 0$. Taking base- e on both sides, we now know $\mu(x)$:

$$\mu(x) = e^{\int f(x) dx}.$$

So let's go back to the start: we have a differential equation of the form

$$y' + f(x)y = g(x).$$

We have found $\mu(x)$ and multiplied it to both sides:

$$\mu(x)y' + \mu(x)f(x)y = \mu(x)g(x).$$

The left side is compressed as a product rule derivative:

$$(\mu(x)y)' = \mu(x)g(x).$$

Now integrate both sides. Here, we need the constant of integration:

$$\int (\mu(x)y)' = \int \mu(x)g(x)dx + C.$$

Since $\int (\mu(x)y)' = \mu(x)y$, we can now solve for y :

$$y = \frac{\int \mu(x)g(x)dx + C}{\mu(x)}.$$

You may memorize this as a formula.

Example: Find the general solution of $y' + \frac{2}{x}y = x$.

(Note that $f(x) = \frac{2}{x}$ and $g(x) = x$ and that $x \neq 0$)

Solution: First, we find $\mu(x)$:

$$\mu(x) = e^{\int \left(\frac{2}{x}\right) dx} = e^{2 \ln x} = e^{\ln x^2} = x^2.$$

Now, use the formula $y = \frac{\int \mu(x)g(x)dx + C}{\mu(x)}$: *(next slide)*

Solve $y' + \frac{2}{x}y = x$. From previous slide, we know that $\mu(x) = x^2$.

Now we use the formula $y = \frac{\int \mu(x)g(x)dx + C}{\mu(x)}$:

$$y = \frac{\int x^2 x \, dx + C}{x^2} = \frac{\int x^3 \, dx + C}{x^2} = \frac{\frac{1}{4}x^4 + C}{x^2} = \frac{1}{4}x^2 + Cx^{-2}.$$

Thus, $y = \frac{1}{4}x^2 + Cx^{-2}$ is the general solution of $y' + \frac{2}{x}y = x$.

Check that $y = \frac{1}{4}x^2 + Cx^{-2}$ is the general solution of $y' + \frac{2}{x}y = x$. First, differentiate y :

$$y' = \frac{1}{2}x - 2Cx^{-3}$$

Now insert y' and y into the differential equation and simplify:

$$\begin{aligned}\left(\frac{1}{2}x - 2Cx^{-3}\right) + \frac{2}{x}\left(\frac{1}{4}x^2 + Cx^{-2}\right) &= x \\ \left(\frac{1}{2}x - 2Cx^{-3}\right) + \left(\frac{1}{2}x + 2Cx^{-3}\right) &= x \\ \left(\frac{1}{2}x + \frac{1}{2}x\right) + (-2Cx^{-3} + 2Cx^{-3}) &= x \\ x + 0 &= x.\end{aligned}$$

Example: Solve the IVP: $y' + 2xy = 3x$, $y(0) = 5$.

$$\{\mu(x) = e^{\int 2x dx} = e^{x^2}\}:$$

Now find y using the formula $y = \frac{\int \mu(x)g(x)dx + C}{\mu(x)}$:

$$y = \frac{\int \mu(x)g(x)dx + C}{\mu(x)} = \frac{\int e^{x^2}(3x)dx + C}{e^{x^2}} = \frac{\frac{3}{2}e^{x^2} + C}{e^{x^2}} = \frac{3}{2} + Ce^{-x^2}.$$

$$\text{To find } C, \text{ use } y(0) = 5: 5 = \frac{3}{2} + Ce^{-(0)^2} \rightarrow C = \frac{7}{2}.$$

Thus, the particular solution is

$$y = \frac{3}{2} + \frac{7}{2}e^{-x^2}.$$